Description:
- Template: ./src/experiments/experiments_template.py

SHARED
Experiment Template

preprocess-flag
Preprocessing data

- Preprocess and save splits under: ./data/preprocessed/[setting]_splits
- Naming Convention: [setting_type.csv], e.g. one_shot_train.csv

train-flag

config.py, data.py

SHARED
Experiment Metadata

Description: create_experiment_dir()
- create experiment_id from experiment parameters
- create experiment folder

[folder: e.g. one_shot__EN__target_input_variant__mBERT__seed51/]

SHARED
Set seeds & configure environment

Description:
- set seeds → set_seeds()
- select device
- initialize ModelRunner()

SHARED
Load data and modify inputs

Description: load_data_splits(), build_inputs_for_splits(), compute_and_check_split_stats()
- get data corresponding to experiment config
- build Inputs according to experiment config
- create split stats and check sample size (split_stats.csv)
- Input: Experiment_config and paths
- Output: train_data, dev_data, test_data

training.py, models

SHARED
Check model checkpoints

Description: get_model()
- check for pretrained model for experiment config
- Skip if full artifacts exist otherwise train
- Input: experiment_config, experiment_dir, train_data, val_data, runner
- Output: model, best_params

no

SPECIFIC
Hyperparameter Tuning

Description:
- **ModelRunner.tune()**
  - set seeds
  - calls inside tuning which runs over the individual hyperparameter grid (in experiment_config)
- Parameter Grid: ./src/models/[model]/param_grid.py

- Input: experiment_config, model_path, train_data, dev_data
- Output: model, tuning_results, best_params, best_curves
  - Trained Model (early stopping on val)
  - Saved model checkpoints [model.pth] (early stopping on dev)
  - Saved tuning_results.csv/.json (dev metric for each hyperparameter)
  - Saved best_params.json
  - Saved loss curves as learning_curves.json

yes

SPECIFIC
Prepare inputs

Description:
- **ModelRunner.prepare_features()**
- Input: train_data, dev_data, experiment_config, best_params
- Output:
  - tokenized/featurized data

SPECIFIC
Initialize model

Description:
- **ModelRunner.initialize()**
- Input: best_params, seed, model_family
- Output: Initialized model according to experiment config

SPECIFIC
Load model checkpoint

Description: load_model()
- **ModelRunner.initialize()**
- Input: best_params, seed, model_family
- Output: loaded model from checkpoints

SPECIFIC
Predict on test

Description:
- **ModelRunner.predict_proba()**
  - Output: y_proba
- Input: loaded model from checkpoints, test data
- Output: test_predictions (id, y_proba, label, y_pred, mwe)

evaluation.py

SHARED
Compute metrics

Description: compute_metrics()
- Input: y_pred, label
- Output:
  - computed macro Recall & Precision
  - computed macro F1 (per label)

SHARED
Save artifacts

Description: save_artifacts()
- save test_predictions.csv/.json
- save metrics.csv/.json (macro recall, macro precision, macro F1)
- save experiment_config.json

SHARED
Evaluation

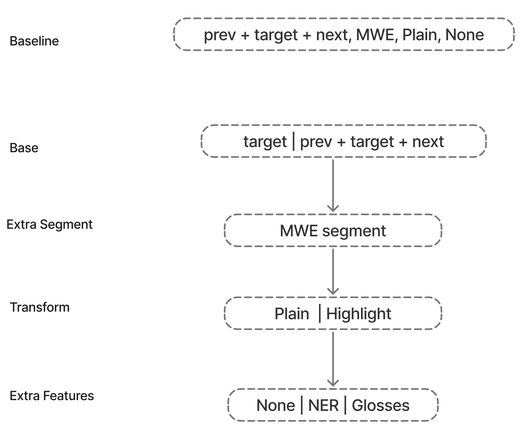
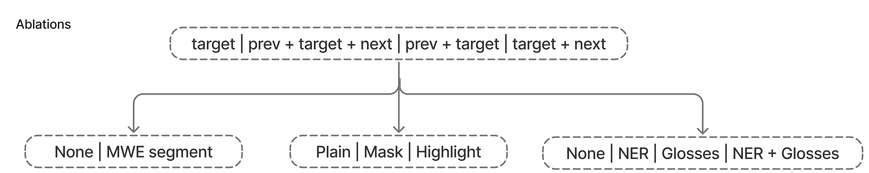
evaluate-flag

SPECIFIC SHARED
Analysis

analysis-flag

analysis, analysis_submodule

Input variants used



Experiments